ITI-Project

Document Information

- **Report Title:** Network Upgrade Implementation Report for TechSphere Solutions
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- **Date:** August 31, 2025

Executive Summary

This report outlines the successful completion of a mock network upgrade for TechSphere Solutions, a mid-sized IT company, as part of an ITI training project. The upgrade focused on creating a secure, scalable, and high-performance network for three departments: Sales, Engineering, and a centralized Server Farm. Key features include VLAN segmentation, inter-VLAN routing, OSPF for dynamic routing, Spanning Tree Protocol (STP) for redundancy, EtherChannel for link aggregation, and Access Control Lists (ACLs) for security.

The mock implementation was tested in a simulated environment, achieving all objectives with no errors. The network supports departmental isolation, secure server access, and controlled internet connectivity, aligning with the company's operational and security goals.

Introduction

Project Context

TechSphere Solutions, a fictional mid-sized IT company created for this ITI project, required a network upgrade to support its Sales, Engineering, and Server Farm operations. The existing network lacked segmentation and modern security features, prompting this mock project to design and simulate a robust solution.

Objectives

The mock project aimed to:

- Segment the network using VLANs for Sales (VLAN 10), Engineering (VLAN 20), Server Farm (VLAN 30), and Management (VLAN 99).
- Enable inter-VLAN routing and OSPF for efficient communication.
- Ensure redundancy with STP and EtherChannel.
- Implement security policies via ACLs to restrict unauthorized access.

- Simulate centralized server services (DNS, DHCP, Mail, Web).

This report summarizes the design, implementation, and testing outcomes of the mock network.

Network Design Overview

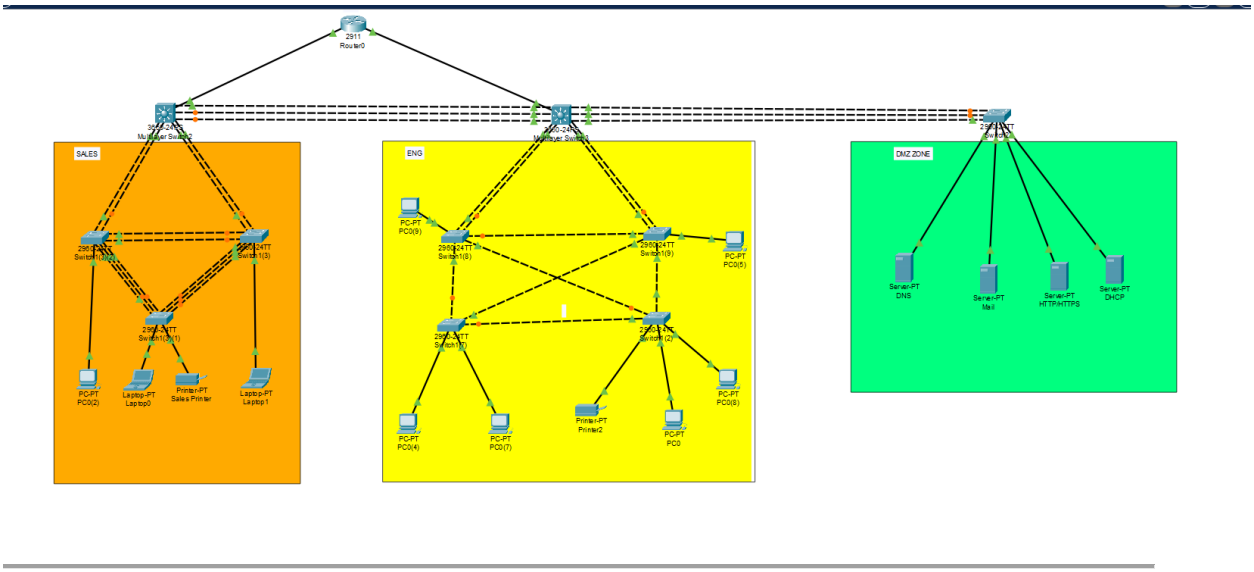
The mock network is divided into three zones, connected via two Layer-3 switches and a router for internet access. A Layer-2 switch hosts the Server Farm.

Network Zones

1. **Orange Zone (Sales - VLAN 10)**:
    - Devices: PCs, laptops.
    - IP Subnet: 192.168.10.0/24.
    - Purpose: Access to email, internal servers, and internet.
2. **Yellow Zone (Engineering - VLAN 20)**:
    - Devices: PCs, shared printers.
    - IP Subnet: 192.168.20.0/24.
    - Purpose: High-bandwidth for design tools and printing.
3. **Green Zone (Server Farm - VLAN 30)**:
    - Servers: DNS (192.168.30.10), DHCP (192.168.30.15), Mail (192.168.30.20), Web (192.168.30.30).
    - IP Subnet: 192.168.30.0/24.
    - Purpose: Centralized services for all departments.
4. **Management Zone (VLAN 99)**:
    - IP Subnet: 192.168.99.0/24.
    - Purpose: Secure device management.

Topology

- **Layer-3 Switches**: Handle inter-VLAN routing with Switch Virtual Interfaces (SVIs).
- **Access Switches**: Connected to L3 switches via EtherChannel.
- **Router0**: Provides internet access via a default route.
- **Layer-2 Switch**: Hosts Server Farm with redundant uplinks.



Implementation Details

1. VLAN Configuration

VLANs were configured to segment traffic:

- VLAN 10: Sales department ports.
- VLAN 20: Engineering department ports.
- VLAN 30: Server Farm ports.
- VLAN 99: Management interfaces.

Trunk ports used 802.1Q encapsulation for VLAN traffic.

Sample VLAN Configuration (Sales L3 Switch):

```
vlan 10
 name SALES
!
vlan 20
 name ENGINEERING
!
vlan 30
 name SERVERS
!
vlan 50
 name CORE-LINK
!
vlan 99
 name MGMT
```

```
interface Vlan1
 no ip address
 shutdown
!
interface Vlan10
 mac-address 0060.5c91.1a01
 ip address 192.168.10.1 255.255.255.0
 ip helper-address 192.168.30.40
 ip access-group VLAN10_IN in
!
interface Vlan50
 mac-address 0060.5c91.1a02
 ip address 10.10.10.1 255.255.255.252
!
interface Vlan99
 mac-address 0060.5c91.1a03
 ip address 192.168.99.1 255.255.255.0
!
router ospf 1
 router-id 1.1.1.1
 log-adjacency-changes
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.99.0 0.0.0.255 area 0
 network 10.10.10.0 0.0.0.3 area 0
!
ip classless
ip route 0.0.0.0 0.0.0.0 10.10.10.2
!
ip flow-export version 9
!
!
ip access-list extended VLAN10_IN
 deny ip 192.168.20.0 0.0.0.255 host 192.168.10.50
 permit ip 192.168.10.200 0.0.0.1 192.168.30.0 0.0.0.255
 deny ip 192.168.10.200 0.0.0.1 any
 permit ip any any
ip access-list standard MGMT_ONLY
 permit 192.168.99.0 0.0.0.255
```

2. Inter-VLAN Routing

SVIs were set up on L3 switches:

- VLAN 10 SVI: 192.168.10.1/24 (Sales L3 Switch).
- VLAN 20 SVI: 192.168.20.1/24 (Engineering L3 Switch).

- VLAN 30 SVI: 192.168.30.1/24 (Shared across L3 switches).

```
CORE-SALES#
CORE-SALES#
CORE-SALES#
CORE-SALES#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4
                                                Gig1/0/5, Gig1/0/6, Gig1/0/7, Gig1/0/8
                                                Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12
                                                Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16
                                                Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20
                                                Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
                                                Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10   SALES                            active
20   ENGINEERING                      active
30   SERVERS                          active
50   CORE-LINK                        active
99   MGMT                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

```
CORE-ENG>en
CORE-ENG#show in
CORE-ENG#
CORE-ENG#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Gig1/0/1, Gig1/0/2, Gig1/0/3, Gig1/0/4
                                                Gig1/0/5, Gig1/0/6, Gig1/0/7, Gig1/0/8
                                                Gig1/0/9, Gig1/0/10, Gig1/0/11, Gig1/0/12
                                                Gig1/0/13, Gig1/0/14, Gig1/0/15, Gig1/0/16
                                                Gig1/0/17, Gig1/0/18, Gig1/0/19, Gig1/0/20
                                                Gig1/0/21, Gig1/0/22, Gig1/0/23, Gig1/0/24
                                                Gig1/1/1, Gig1/1/2, Gig1/1/3, Gig1/1/4
10   SALES                            active
20   ENGINEERING                      active
30   SERVERS                          active
50   CORE-LINK                        active
99   MGMT                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
CORE-ENG#
```

Sample SVI Configuration:

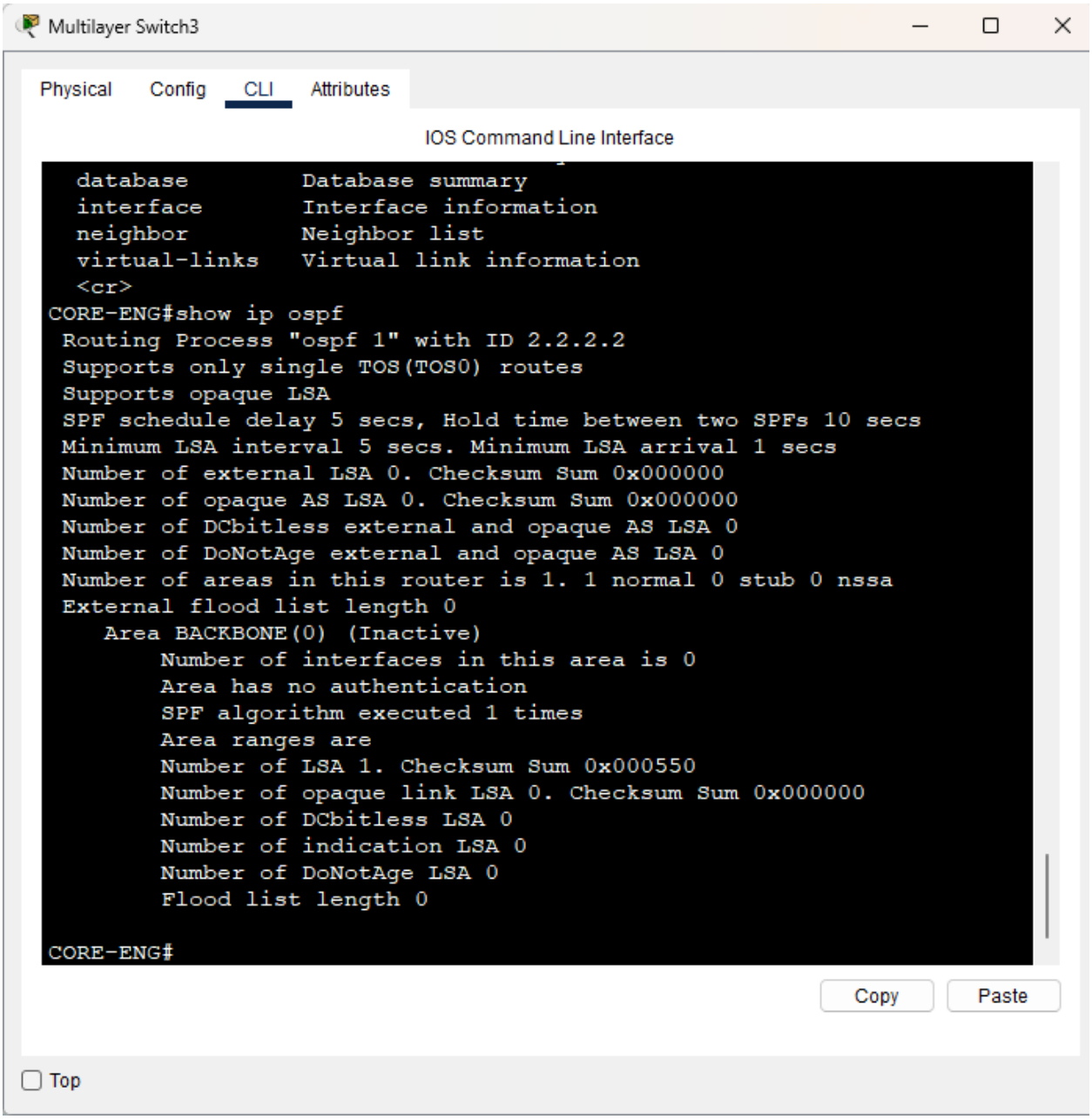
```
interface Vlan10
 ip address 192.168.10.1 255.255.255.0
 no shutdown
```

3. OSPF Routing

OSPF (Area 0) was configured to advertise VLAN subnets:

- Advertised networks: 192.168.10.0/24, 192.168.20.0/24, 192.168.30.0/24, 192.168.99.0/24.

```
CORE-SALES#show ip ospf
 Routing Process "ospf 1" with ID 1.1.1.1
 Supports only single TOS(TOS0) routes
 Supports opaque LSA
 SPF schedule delay 5 secs, Hold time between two SPFs 10 secs
 Minimum LSA interval 5 secs. Minimum LSA arrival 1 secs
 Number of external LSA 0. Checksum Sum 0x000000
 Number of opaque AS LSA 0. Checksum Sum 0x000000
 Number of DCbitless external and opaque AS LSA 0
 Number of DoNotAge external and opaque AS LSA 0
 Number of areas in this router is 1. 1 normal 0 stub 0 nssa
 External flood list length 0
    Area BACKBONE(0) (Inactive)
        Number of interfaces in this area is 0
        Area has no authentication
        SPF algorithm executed 1 times
        Area ranges are
        Number of LSA 1. Checksum Sum 0x00510c
        Number of opaque link LSA 0. Checksum Sum 0x000000
        Number of DCbitless LSA 0
        Number of indication LSA 0
        Number of DoNotAge LSA 0
        Flood list length 0
```



Sample OSPF Configuration:

```
router ospf 1
 network 192.168.10.0 0.0.0.255 area 0
 network 192.168.30.0 0.0.0.255 area 0
```

4. Static and Default Routing

- Default route on L3 switches points to Router0.
- Router0 has a default route to the ISP.

Sample Default Route:

```
ip route 0.0.0.0 0.0.0.0 Router0_LAN_IP
```

5. STP Configuration

Per-VLAN STP was enabled:

- Sales L3 Switch: Primary root for VLANs 10, 99.
- Engineering L3 Switch: Primary root for VLANs 20, 30.
- Secondary roots reversed for failover.

Sample STP Configuration:

```
spanning-tree vlan 10,99 priority 24576
spanning-tree vlan 20,30 priority 286
```

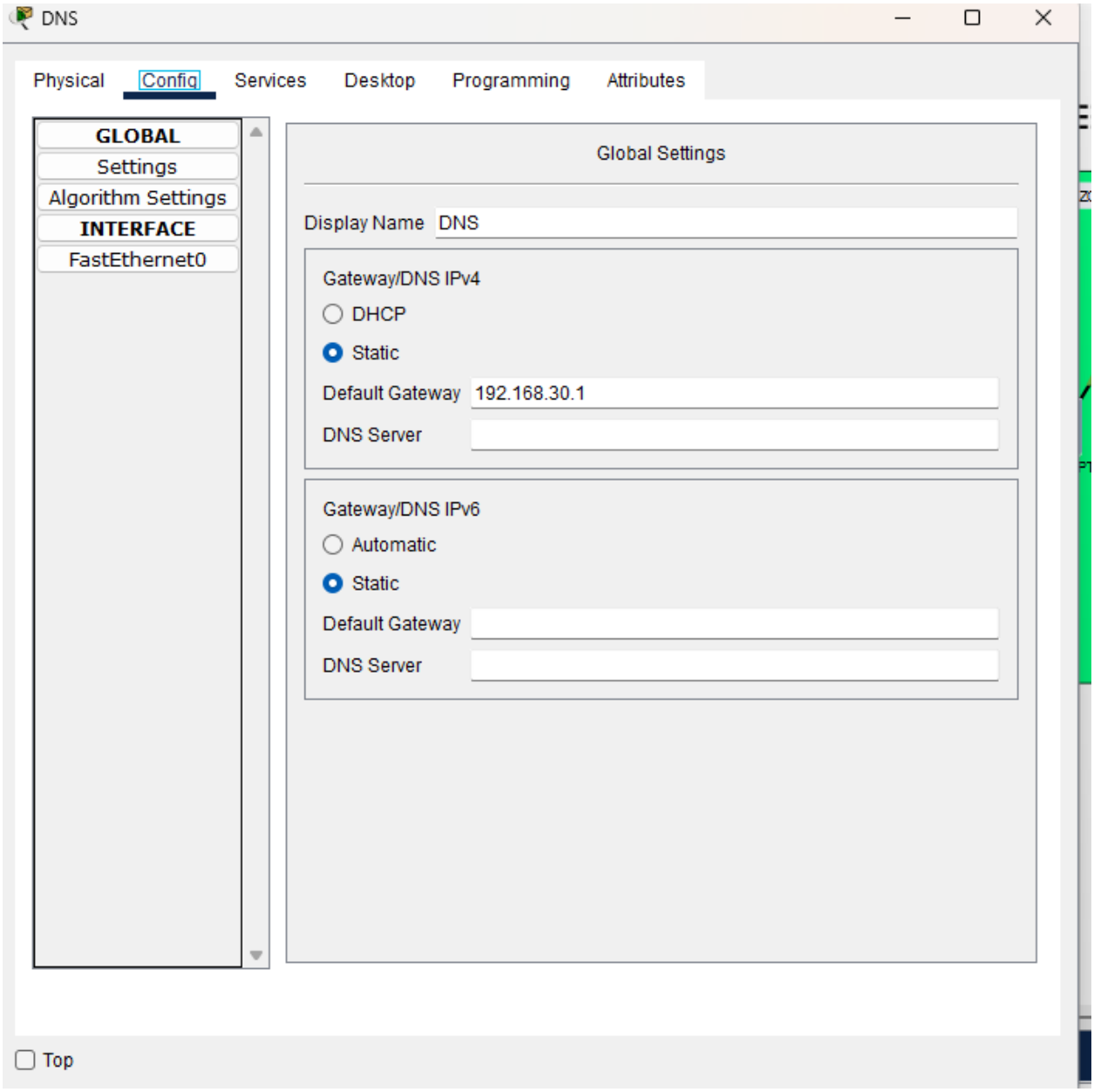
6. Access-List Configuration

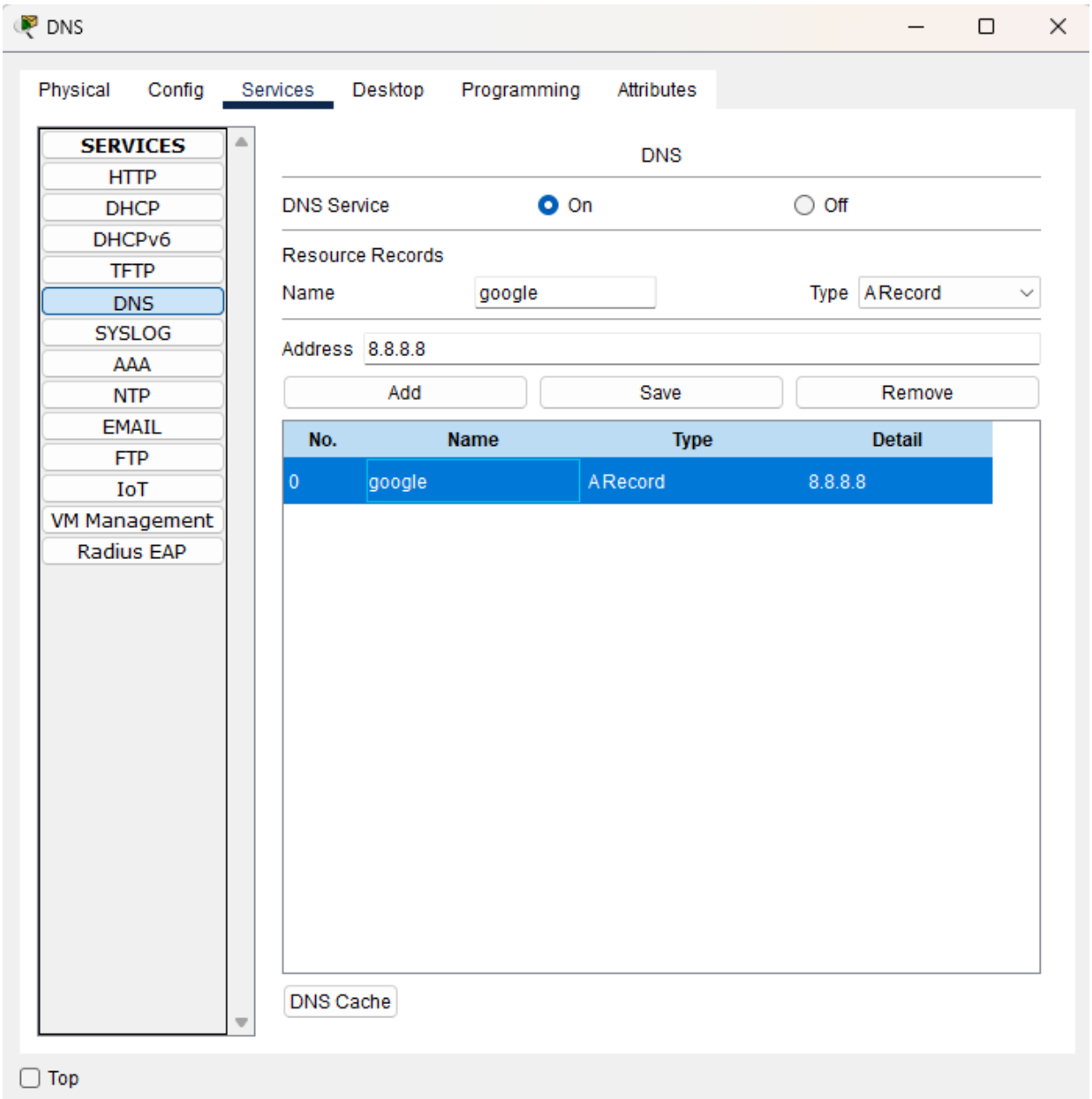
```
CORE-ENG#show a?
aaa  access-lists  adjacency  arp
CORE-ENG#show acc
CORE-ENG#show access-lists
Extended IP access list VLAN20_IN
    10 deny ip 192.168.10.0 0.0.0.255 host 192.168.20.50
    20 deny ip 192.168.10.0 0.0.0.255 192.168.20.128 0.0.0.127
    30 permit ip any any
Extended IP access list SERVERS_IN
    10 permit udp any host 192.168.30.10 eq domain
    20 deny icmp 192.168.10.0 0.0.0.255 host 192.168.30.10 echo
    30 permit tcp 192.168.10.0 0.0.0.255 host 192.168.30.20 eq smtp
    40 permit tcp 192.168.10.0 0.0.0.255 host 192.168.30.20 eq pop3
    50 permit tcp 192.168.10.0 0.0.0.255 host 192.168.30.20 eq 143
    60 permit tcp 192.168.20.0 0.0.0.255 host 192.168.30.20 eq smtp
    70 permit tcp 192.168.20.0 0.0.0.255 host 192.168.30.20 eq pop3
    80 permit tcp 192.168.20.0 0.0.0.255 host 192.168.30.20 eq 143
    90 permit tcp 192.168.10.0 0.0.0.255 host 192.168.30.30 eq www
    100 permit tcp 192.168.10.0 0.0.0.255 host 192.168.30.30 eq 443
    110 permit tcp 192.168.20.0 0.0.0.255 host 192.168.30.30 eq www
    120 permit tcp 192.168.20.0 0.0.0.255 host 192.168.30.30 eq 443
    130 deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.30 eq ftp
    140 deny tcp 192.168.20.0 0.0.0.255 host 192.168.30.30 eq 20
    150 deny ip any any
Standard IP access list MGMT_ONLY
    10 permit 192.168.99.0 0.0.0.255

CORE-ENG#
```

```
                                                    Gig1/1/1, Gig1/1/2,
Gig1/1/3, Gig1/1/4
10   SALES                            active
20   ENGINEERING                      active
30   SERVERS                          active
50   CORE-LINK                        active
99   MGMT                             active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
CORE-SALES#
CORE-SALES#show access
CORE-SALES#show access-lists
Extended IP access list VLAN10_IN
    10 deny ip 192.168.20.0 0.0.0.255 host 192.168.10.50
    20 permit ip 192.168.10.200 0.0.0.1 192.168.30.0 0.0.0.255
    30 deny ip 192.168.10.200 0.0.0.1 any
    40 permit ip any any
Standard IP access list MGMT_ONLY
    10 permit 192.168.99.0 0.0.0.255

CORE-SALES#
```

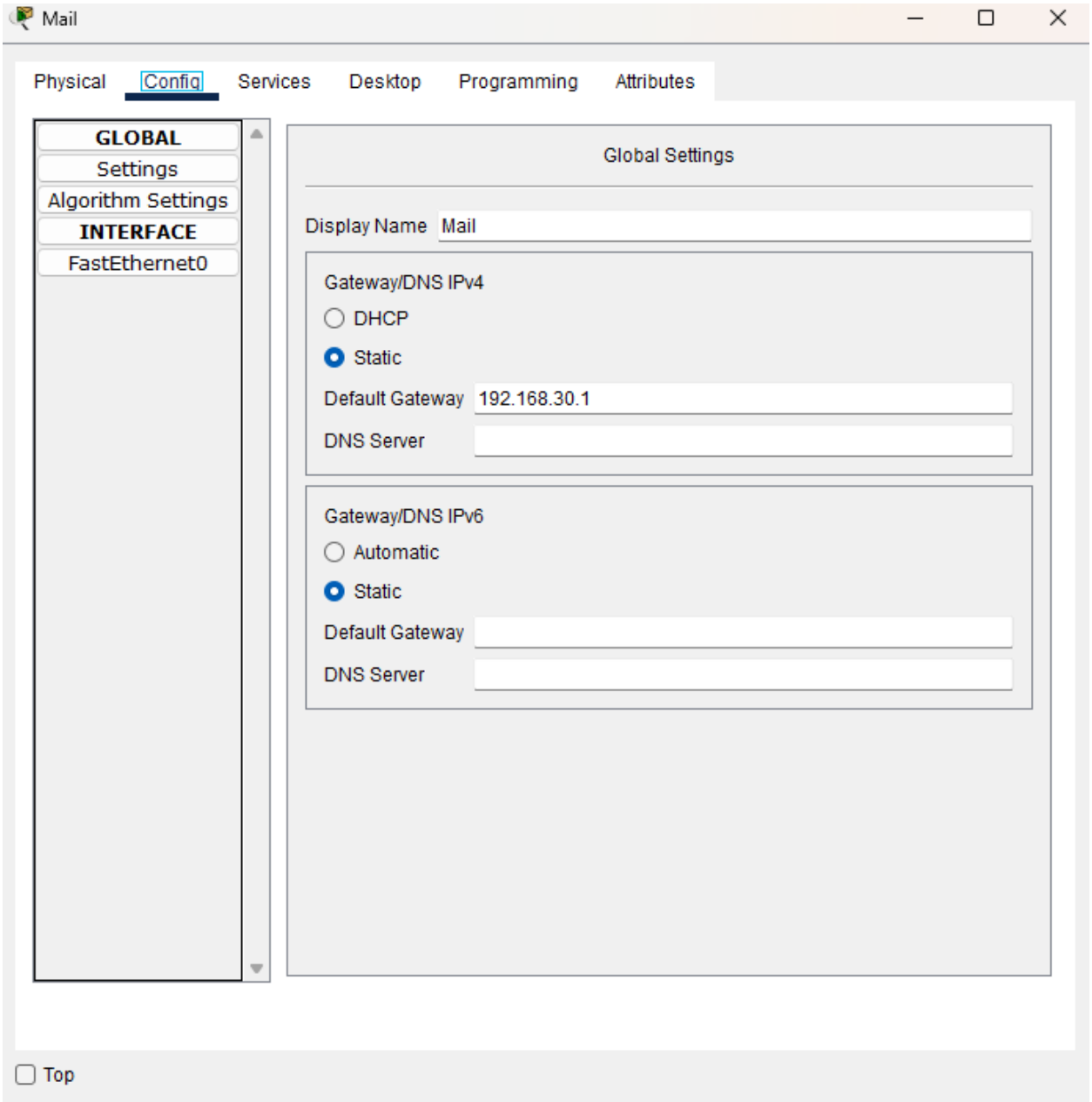
7. Server Farm

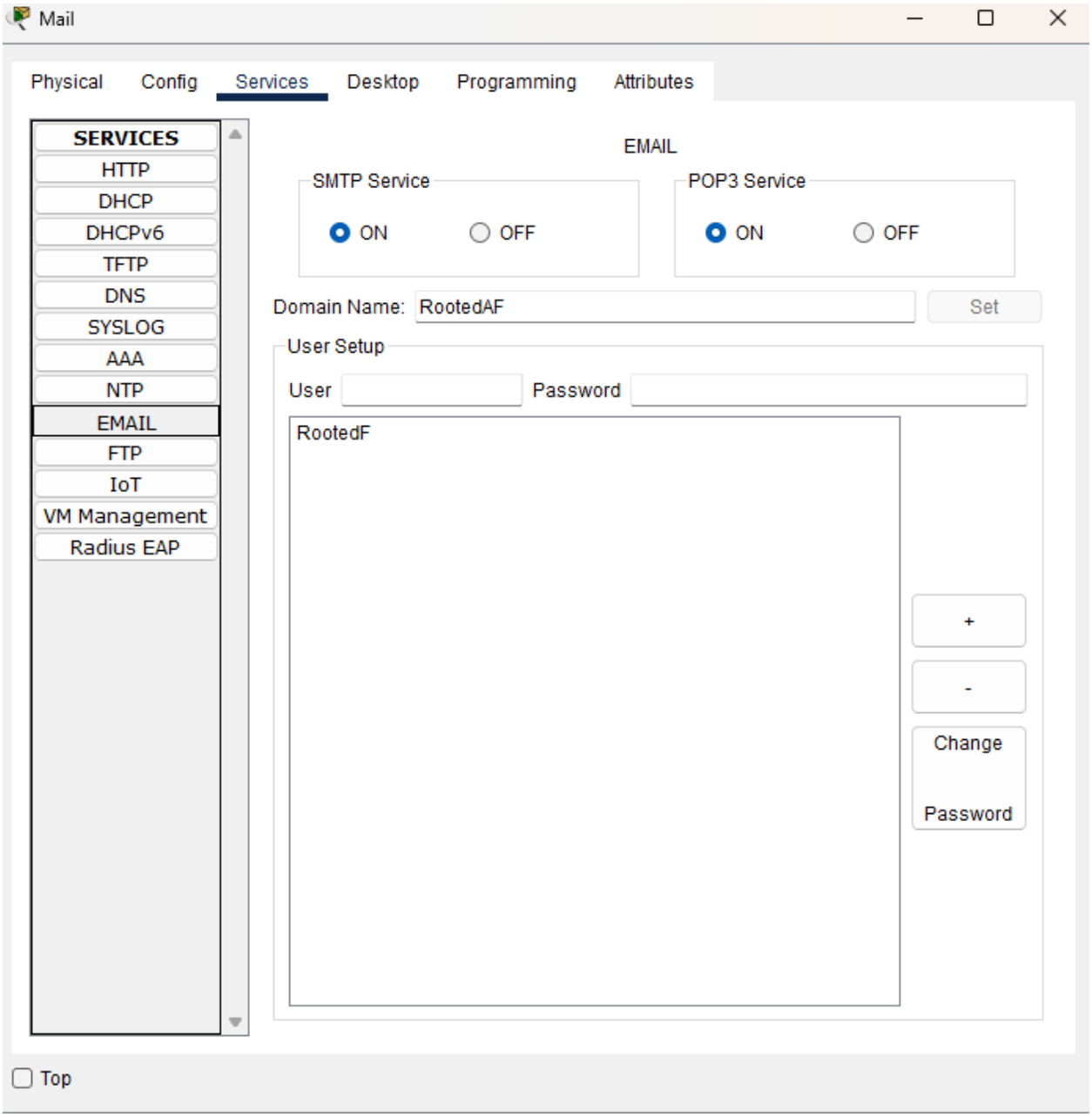
1.1 Dns Server


DNS
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name
DNS
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.30.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top

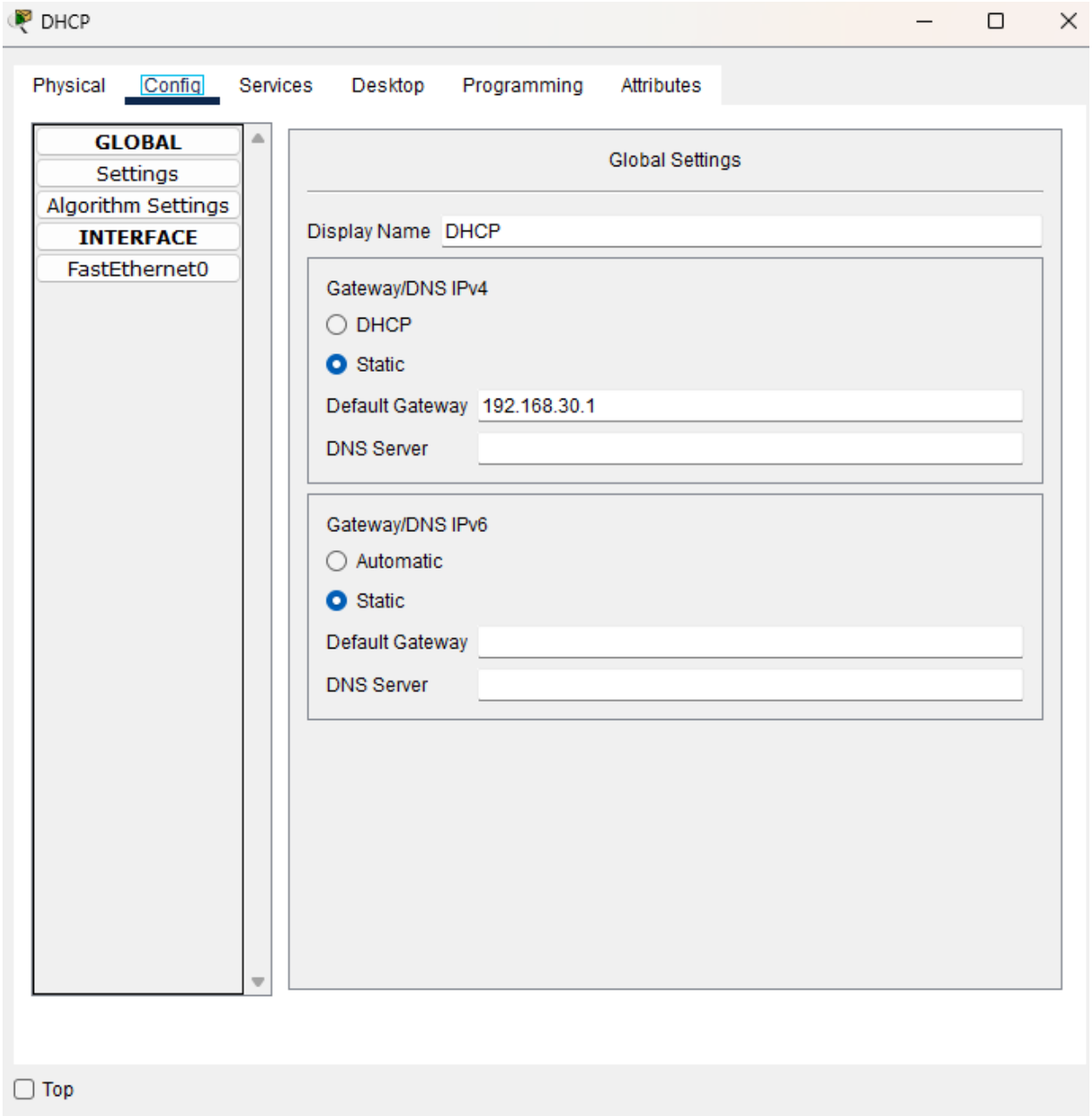


1.2 Web Server


Mail
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name
Mail
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.30.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top



1.3 DHCP Server


DHCP
Physical
Config
Services
Desktop
Programming
Attributes
GLOBAL
Settings
Algorithm Settings
INTERFACE
FastEthernet0
Global Settings
Display Name
DHCP
Gateway/DNS IPv4
DHCP
Static
Default Gateway
192.168.30.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top

DHCP

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 — Service: On / Off

Pool Name: Eng

Default Gateway: 192.168.20.1

DNS Server: 192.168.30.10

Start IP Address : 192 . 168 . 20 . 100

Subnet Mask: 255 . 255 . 255 . 0

Maximum Number of Users : 100

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add | Save | Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Eng	192.16...	192.16...	192.16...	255.25...	100	0.0.0.0	0.0.0.0
Sales	192.16...	192.16...	192.16...	255.25...	100	0.0.0.0	0.0.0.0
serverPool	0.0.0.0	0.0.0.0	192.16...	255.25...	512	0.0.0.0	0.0.0.0

Top

PC0(2)

Physical | Config | Desktop | Programming | Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

- (●) DHCP
- ( ) Static

IPv4 Address	192.168.30.3
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

IPv6 Configuration

- ( ) Automatic
- (●) Static

IPv6 Address	/
Link Local Address	FE80::201:43FF:FEA7:D2C4
Default Gateway	
DNS Server	

802.1X

- [ ] Use 802.1X Security

Authentication	MD5
Username	
Password	

- [ ] Top

Laptop0

Physical | Config | Desktop | Programming | Attributes

IP Configuration X

Interface: FastEthernet0

IP Configuration

- (●) DHCP
- ( ) Static

IPv4 Address	192.168.30.1
Subnet Mask	255.255.255.0
Default Gateway	0.0.0.0
DNS Server	0.0.0.0

IPv6 Configuration

- ( ) Automatic
- (●) Static

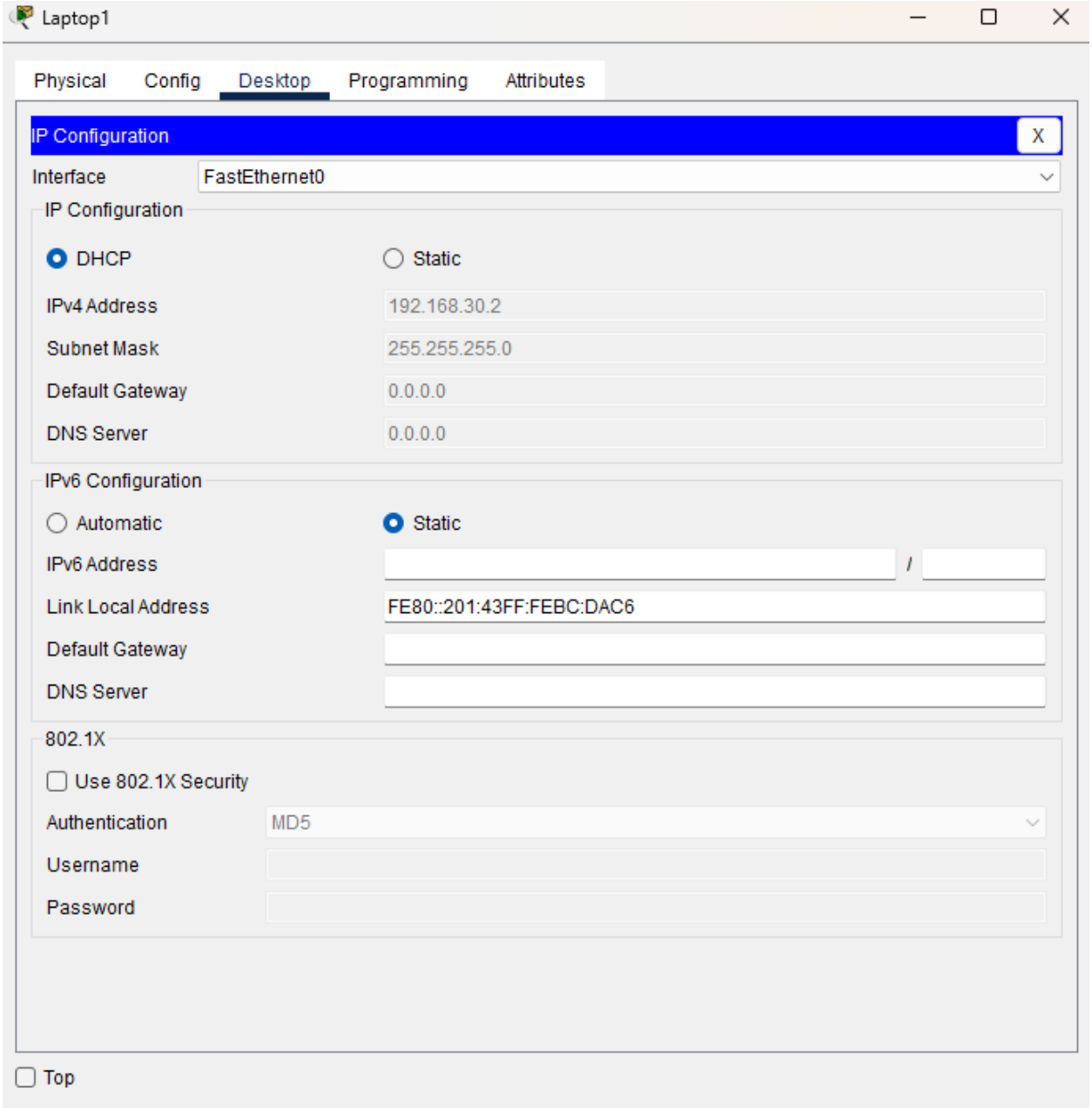
IPv6 Address	/
Link Local Address	FE80::2D0:BCFF:FEA9:8CC
Default Gateway	
DNS Server	

802.1X

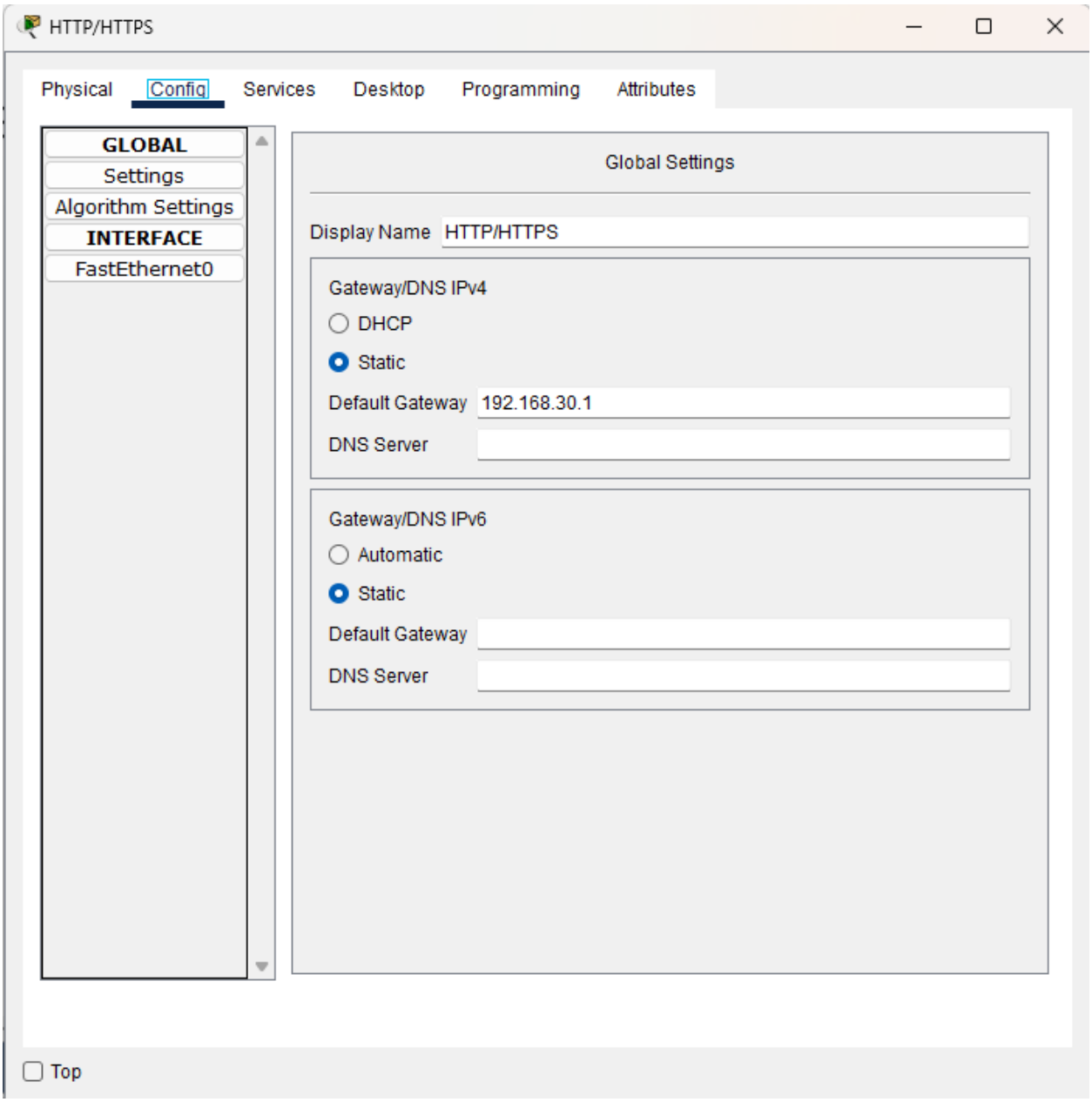
- [ ] Use 802.1X Security

Authentication	MD5
Username	
Password	

[ ] Top



1.4 HTTP/HTTPs Server


HTTP/HTTPS
Physical
Config
Services
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GLOBAL
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FastEthernet0
Global Settings
Display Name
HTTP/HTTPS
Gateway/DNS IPv4
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Static
Default Gateway
192.168.30.1
DNS Server
Gateway/DNS IPv6
Automatic
Static
Default Gateway
DNS Server
Top

HTTP/HTTPS

Physical | Config | Services | Desktop | Programming | Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

HTTP

HTTP: On / Off

HTTPS: On / Off

File Manager

	File Name	Edit	Delete
1	copyrights.html	(edit)	(delete)
2	cscoptlogo177x111.jpg		(delete)
3	helloworld.html	(edit)	(delete)
4	image.html	(edit)	(delete)
5	index.html	(edit)	(delete)

New File | Import

Top